Educational

By:Brian Kaestner

1. **Website Type & Purpose**

I am going to be choosing an educational website. This aligns with my goals as a current educator. I will be looking at how I can better my online teaching through creating an interactive website for students to use during online learning days. This will be an instructional page that will teach a new topic for students to learn at home.

2. **Home Page Content**

- Welcome statement (1- 2 sentences).

  Welcome to our Algebra 2 learning space, where exponential equations become approachable and even understandable. This page will guide you step-by-step as you build confidence with one of the most powerful topics in mathematics solving equations.

- Short paragraph on your topic (100-150 words).).

  Solving exponential equations is a key skill in Algebra 2 and an essential part of understanding how quantities grow and change in the real world. These equations appear in situations involving population growth, radioactive decay, compound interest, and many other natural and financial processes. In this lesson, you'll learn how to recognize exponential equations by identifying growth and decay, and rewrite expressions using common bases to solve equations that can't be simplified by inspection. You'll also explore strategies for isolating exponential expressions and avoiding common mistakes students often make. By the end, you'll have a strong foundation for solving exponential equations with confidence and accuracy. These skills will support you in advanced math courses and real-life applications.

- Background / highlight image

  A clean, modern graphic showing an exponential curve rising across a coordinate plane, with labeled axes and a subtle grid background. The curve should be smooth, bright, and visually appealing—symbolizing growth, change, and the power of exponential relationships.

- a key statement / motto / quote about your topic

  "Small steps lead to big growth — just like exponential functions."

3. **About / Educator Credentials Page**

Hello my name is Brian Kaestner. I am working towards my masters degree at the University of Missouri. I am studying Learning Technology and Design with an emphasis in school. I plan to graduate in December 2026. I have my undergraduate from Lindenwood University in mathematics education. I am currently a high school math teacher but have also taught middle school in the past. Some of my hobbies include camping at different parks such as; The Grand Canyon, Yellowstone, Banff, Jasper, Glacier, Bryce Canyon, and Zion to name a few. I am also interested in pc gaming and woodworking. Most of my woodworking experience is from working on different projects around my house. My wife and I have two kids now, a three year old boy and a one year old baby girl. The two kids now take up most of my time and money.

Kaestner resume.pdf

Movie on 2-1-26 at 11.21 AM.mov

(I don't think it plays yet, before a download. I will be looking into uploading it better later. Right now it's just in my drive.)

4. **Projects/Experience OR Business/ Content Page**

- Understanding and Applying the Change-of-Base Formula

I can explain and apply the change-of-base formula to evaluate each step in solving an exponential equation.

Solving exponential equations using exponent properties | High School Math | Kha…

- Verifying Solutions in Exponential Models

I can check the reasonableness of my solution by substituting it back into the original exponential equation and interpreting the result.

(Image or video of showing this skill.)

- Graphing Exponential Equations

I can graph exponential functions and describe how changes in the equation affect the graph's growth or decay.

Exponential Functions | Desmos

5. Resources

IXL Learning. (n.d.). *Solve exponential equations by rewriting the base*. IXL. https://www.ixl.com/math/algebra-2/solve-exponential-equations-by-rewriting-the-base

Will be used for practice solving equations.

McGraw Hill. (2012). *Algebra 2* (Student ed.). McGraw Hill. https://www.mheducation.com/prek-12/program/microsites/MKTSP-GCA06M0/algebra-2.html

McGraw Hill is the text book that we use at my school, and was used for examples.

Khan Academy. (2016, February 17). *Solving exponential equations using exponent properties* [Video]. YouTube. https://www.youtube.com/watch?v=etl9KKf6se0

This video is for solving exponential equations.

Missouri Department of Elementary and Secondary Education. (2020). *Missouri Learning Standards: Mathematics (Revised 2020)*. https://dese.mo.gov/college-career-readiness/curriculum/missouri-learning-standards

This website is where I got the information for my learning standards and what to pick.

Desmos. (n.d.). *Exponential functions* [Interactive graph]. https://www.desmos.com/calculator/3fisjexbvp

Uses for graphing exponential functions in expiration.

*Microsoft Copilot was used as a digital writing assistant during the development of this material, providing support with idea generation, organization, and wording.